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Short Communication

Microstructure and tensile properties of $FeMnNiCuCoSn_x$ high entropy alloys

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ABSTRACT

High-entropy $FeMnNiCuCoSn_x$ (x denotes the atomic fraction of Sn) alloys with good plasticity have been developed. The systematical investigation demonstrates that the concentration of Sn element plays a significant role in the microstructure and tensile properties. As $0.03 < x < 0.05$, the alloys exhibit high tensile strength and plasticity, and the maximum elongation strain and strength are 16.9% and 476.9 MPa, respectively, because of their single FCC solutions. While the concentration of Sn is higher than 0.05, an intermetallic compound ($Cu_{5.6}Sn$) in the interdendritic regions forms, which degrades the ductility of alloys.

1. Introduction

The technologic improvement in engineering, especially the aerospace industry, usually relies on the development of new structural materials with better performance compared to the traditional ones. In the past decade, the high entropy alloys (HEAs), which include 5–13 principal elements at equimolar or near-equimolar compositions, have been proposed and developed [1–3]. The atomic fractions of each element in HEAs cannot be less than 5% or more than 35% [1], producing the high mixing entropy that prevents the formation of intermetallic phases. Therefore, these alloys predominantly consist of a mixture of simple solid solutions and favorable combination of compression strength and ductility [4–6]. Many HEAs have high hardness [1], strength [7–9] and wear resistance [10], as well as microstructure stability against heat treatment [11–13]. In despite of attractive mechanical and structural features, the low ductility and high brittleness of HEAs seriously limit their potential applications in engineering, especially at room temperature. Although it was demonstrated that the ductility of the cast HEAs could be improved through annealing [14,15], up to now, only several works reported the tensile properties of annealed HEAs (not the cast ones) [16–19]. While almost all works demonstrated the mechanical properties by compression testing or measuring hardness.

To further improve the mechanical properties of HEAs, great efforts have been made by adding a spot of other elements [6,8]. By virtue of the physical features of Sn element, such as low melting temperature of 505.06 K and low solubility with other elements (Fe, Mn, Ni, Cu and Co), it was realized that Sn element could segregate and form a soft phase at grain boundaries, thus improving the plastic properties of HEAs. This was verified by Yang et al. who demonstrated that Sn element is good at improving the Young's modulus and plastic properties of $TiNb_{24}Zr_4Sn_{7.9}$ alloys [20,21].

In this contribution, we report the $FeMnNiCuCoSn_x$ HEA system with different Sn content that was prepared in an arc furnace. The microstructural and mechanical characterizations measured at room temperature demonstrate the significant effect of Sn content on the microstructure and tensile properties.

2. Experimental methods

The $FeMnNiCuCoSn_x$ alloys (x denotes the atomic fraction of Sn, in this paper x = 0, 0.03, 0.05, 0.08, 0.1 and 0.2) were prepared in vacuum arc furnace (WK ΙΙA) under argon atmosphere at preparation temperature of 1800 °C. The purity of all the raw elemental metals was above 99.9%. Table 1 shows the amounts (g) of raw metals in experiments. Repeated melting for at least five times was carried out to improve chemical homogeneity of the alloys. The alloys were then directly solidified in a water-cooled copper mold with a cooling rate similar to 10^3 K/s. The solidified cylindrical rod ingots were about 6 mm in diameter and 70 mm in length. Each tested $FeMnNiCuCoSn_x$ alloys (x = 0, 0.03, 0.05, 0.08, 0.1 and 0.2) was prepared five samples for tensile testing to ensure the accuracy of experiments.

Fig. 1 presents the dimensions (from ISO 6892-1: 2009 [22]) of the samples used in the tensile test. The samples were tested at a crosshead rate of 0.1 mm/min on 810 Material Test System (MTS810). The phase structure analysis of the samples was performed on Rigaku D/max 2500 X-ray diffractometer at 50 kV and 250 mA (XRD, D/Max 2500pc). Scanning angles ranged from 20° to 120° with a scanning rate of 5 deg/min. The microstructure of the alloys was observed by scanning electron microscope (SEM,

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Table 1
The amounts of raw metals in the experiments.

Raw metals	Fe	Mn	Ni	Cu	Co	
Weight (g)	7.66	7.54	8.04	8.72	8.08	
x	0	0.03	0.05	0.08	0.1	0.2
Sn weight	0	0.49	0.82	1.32	1.64	3.28

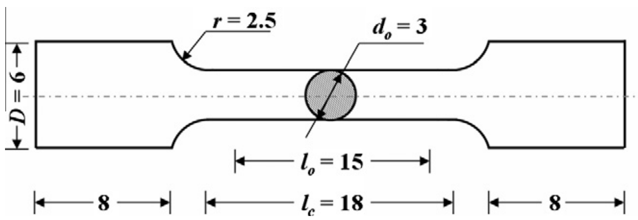

Fig. 1. Dimensions of tensile specimens. Unit: mm.

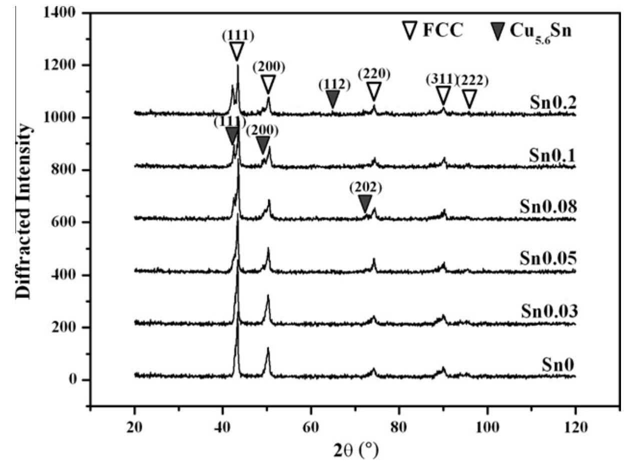

Fig. 2. X-ray diffraction curves of the FeMnNiCuCoSn$_x$ alloys with different Sn contents.

EVO18) and transmission electron microscope (TEM, JEM-2100F). The selected area composition analysis of the samples was carried out by an energy dispersive spectrometry (EDS, GENESIS2000XMS60). Vickers micro-hardness (HV) was measured on polished cross-section surfaces using a 136° vickers diamond pyramid under a 250 g load and 15 s applied time.

3. Results and discussion

Fig. 2 shows the X-ray diffractions (XRDs) of the as-casted FeMnNiCuCoSn$_x$ alloys with different Sn concentrations. As we can see, the FeMnNiCuCoSn$_x$ alloys possess single FCC structure with $0 < x < 0.05$. When x further increases from 0.05, a new intermetallic compound, Cu$_{5.6}$Sn, appears. Obviously, the microstructures of the FeMnNiCuCoSn$_x$ alloys are relatively simple compared to those of the traditional HEAs, especially for the ones with Mn element, in which many intermetallic compounds were demonstrated [14,23,7,24,25]. Nevertheless, there is not any Mn-involved intermetallic compound in our alloy system, implying that all Mn atoms should dissolve into the solid solutions. Since the price of Mn is much lower than that of Ni, Cu, and Co, the addition of Mn element can significantly reduce the cost.

Fig. 3 presents the top-view SEM images of the as-casted FeMnNiCuCoSn$_x$ alloys, showing the dramatical change for alloys with different x. When x is less than 0.05, the microstructures of the as-casted alloys are typical dendritic crystal structures. As x increases to 0.05, Cu$_{5.6}$Sn intermetallic compound appears, and the grain boundaries become increasingly wider because of the growth of the Cu$_{5.6}$Sn particles at the interdendritic regions, as shown in Fig. 4a. With the further addition of Sn ($x = 0.1$), more Cu$_{5.6}$Sn particles form at the interdendritic regions and connect with each other [Fig. 4b]. At $x = 0.2$, Cu$_{5.6}$Sn blocks in interdendritic regions, as shown in Fig. 4c. The volume fraction of Cu$_{5.6}$Sn in the as-casted FeMnNiCuCoSn$_x$ alloys with different x can be roughly calculated by following equation [26]:

$$W_{pi} = \left(p \Big/ \sum p_i\right) \times 100\% \quad (1)$$

where w_{pi} is the volume fraction of the ith phase, p is the peak intensity of a given phase in the XRD pattern, i is the number of the phase, and $\sum p_i$ is the total peak intensity of all phases in the XRD pattern, respectively. The calculated volume fractions of Cu$_{5.6}$Sn and FCC solid solutions are presented in Table 2.

Fig. 5 shows the TEM micrographs of the FeMnNiCuCoSn$_{0.03}$ and FeMnNiCuCoSn$_{0.05}$ alloys. It can be seen that in the FeMnNiCuCoSn$_{0.05}$ alloy, there are massive precipitates which are determined as Cu$_{5.6}$Sn intermetallic compound according to the selected area electron diffraction (SAED) pattern (inset of Fig. 5). The existence of hard and brittle Cu$_{5.6}$Sn gives rise to the decrease of the tensile strength of FeMnNiCuCoSn$_{0.05}$ alloy. In contrast, for the case of the FeMnNiCuCoSn$_{0.03}$ alloy, less addition of Sn does not produce any intermetallic compounds instead of the twin crystals and lots of dislocations, as clearly shown in Fig. 5a, which can improve the mechanical properties of the alloy [27–29].

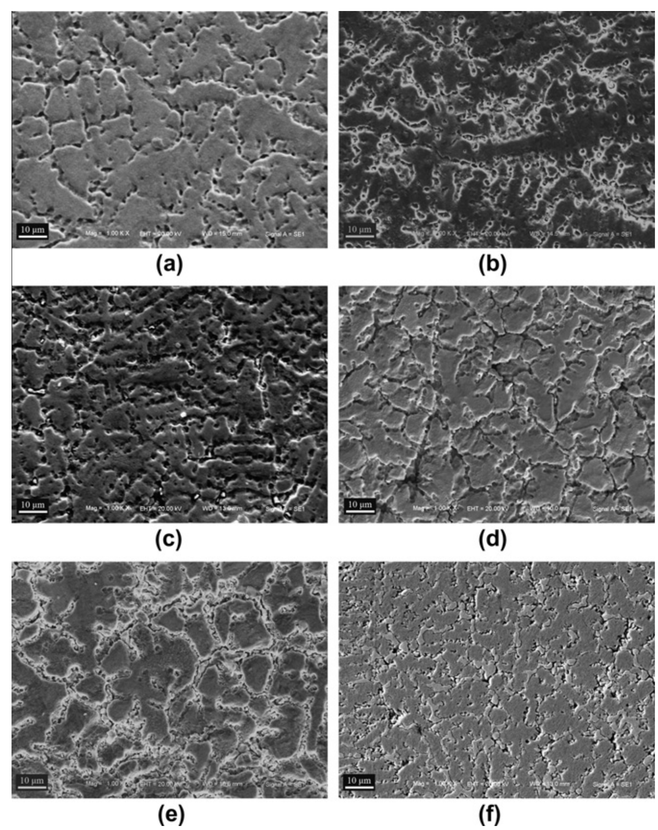

Fig. 3. SEM micrographs of FeMnNiCuCoSn$_x$ alloys with different Sn contents: (a) $x = 0$, (b) $x = 0.03$, (c) $x = 0.05$, (d) $x = 0.08$, (e) $x = 0.1$, and (f) $x = 0.2$.

Table 3 gives the element distributions of the as-casted FeMnNiCuCoSn$_x$ alloys determined by EDS, indicating that the

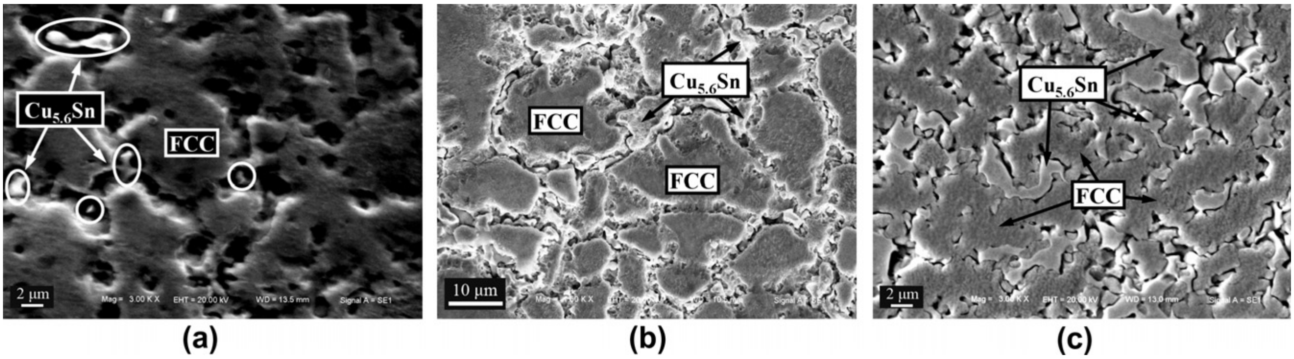

Fig. 4. Magnified SEM micrographs of the $FeMnNiCuCoSn_x$ alloys: (a) $x = 0.05$, (b) $x = 0.1$, and (c) $x = 0.2$.

Table 2
Volume fraction of each phase in as-cast alloys (%).

x	0	0.03	0.05	0.08	0.1	0.2
FCC	100	100	84.4	79.9	67.4	61.9
$Cu_{5.6}Sn$	0	0	15.6	20.1	32.6	38.1

dendritic and indendritic regions are rich in Fe, Co, and Mn, Cu, respectively, while the distribution of Ni is more uniform in the whole specimens. Moreover, it can be found that Cu segregates seriously than other components except for Sn due to the small binding energies of Cu with other atoms [30]. It is difficult for Sn atoms to form solid solution because of the large atomic diameter and low solubility with other elements. Therefore, when x is relatively low ($x < 0.05$), Sn segregates in interdendritic regions as a single phase, benefiting for the plasticity. While $x > 0.05$, Sn will combine with Cu to form $Cu_{5.6}Sn$ in the interdendritic regions,

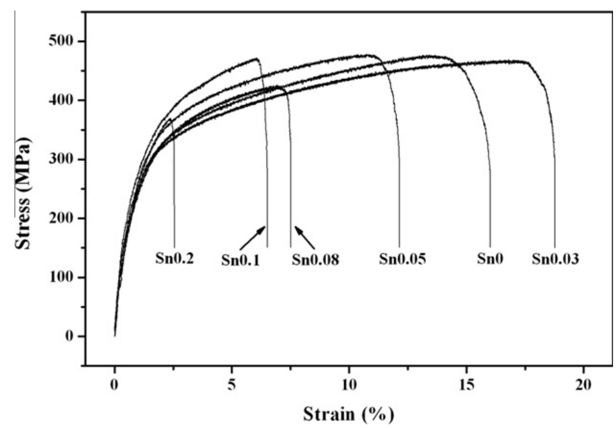

Fig. 6. Tensile curves of the as-cast $FeMnNiCuCoSn_x$ alloys with different Sn contents.

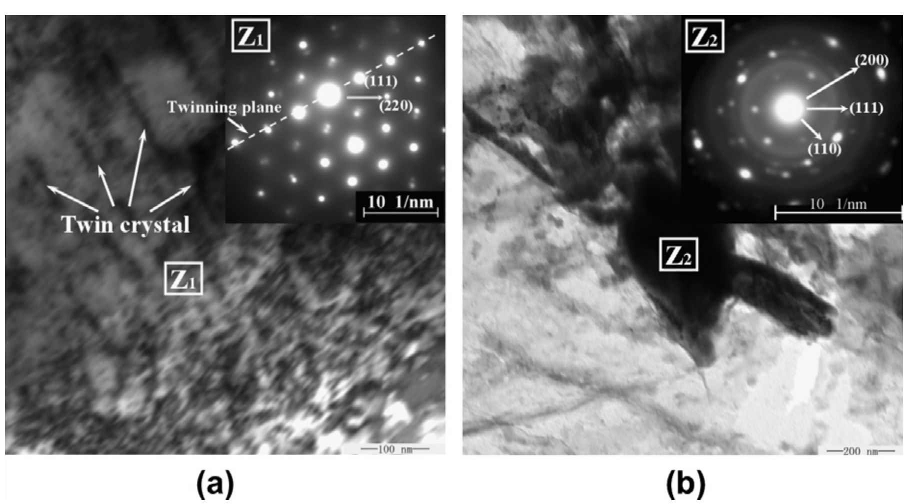

Fig. 5. TEM micrographs of the $FeMnNiCuCoSn_x$ alloys: (a) $FeMnNiCuCoSn_{0.03}$ alloy, (b) $FeMnNiCuCoSn_{0.05}$ alloy.

Table 3
Dendritic (De) and interdendritic (In) compositions determined by EDS (at.%).

Alloy	Mn		Fe		Co		Ni		Cu		Sn	
	De	In	De	In	De	In	De	In	De	In	De	In
$x = 0$	14.08	20.95	31.86	16.36	28.06	17.62	17.83	22.02	8.17	23.05	–	–
$x = 0.03$	14.73	24.51	27.22	10.10	27.55	10.05	20.22	19.18	10.13	33.61	0.15	2.55
$x = 0.05$	16.89	24.34	30.41	10.21	27.11	11.90	17.11	20.88	8.38	28.91	0.11	3.75
$x = 0.08$	12.63	25.4	31.08	3.87	28.29	6.73	18.85	24.73	8.97	28.27	0.18	11.00
$x = 0.1$	14.72	23.57	28.96	7.86	27.41	9.77	19.04	24.12	9.53	24.39	0.33	10.30
$x = 0.2$	10.46	21.41	22.87	13.1	23.31	13.3	21.49	19.38	11.03	23.05	0.85	9.76

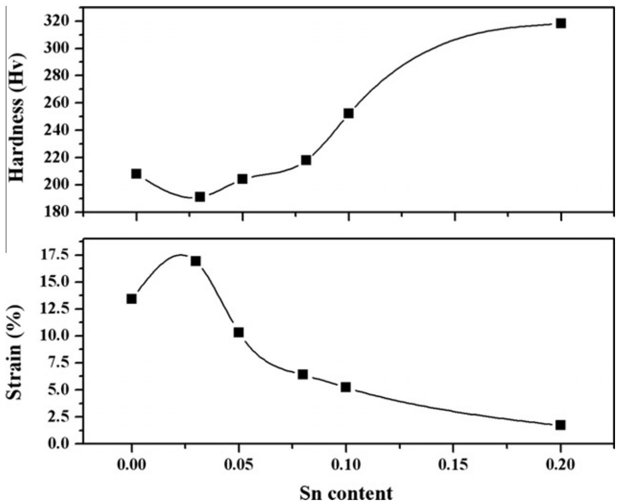

Fig. 7. Tensile strain and the hardness curve of the $FeMnNiCuCoSn_x$ alloys with different Sn contents.

which degrades the ductility. As shown in Fig. 6, the tensile stress–strain curves of the as-cast $FeMnNiCuCoSn_x$ alloys with different x are illustrated. The $FeMnNiCuCoSn_{0.03}$ alloy displays the largest elongation strain of 16.9%, while the $FeMnNiCuCoSn_{0.05}$ alloy displays the highest tensile strength of 476.9 MPa. The $FeMnNiCuCoSn_{0.2}$ alloy has the lowest elongation strain and tensile strength, just 1.7% and 367.7 MPa.

Fig. 7 shows the curve of the tensile strain and hardness as a function of x. As we can see, the tensile strain increases but the hardness decreases for $x < 0.03$, while the tensile strain decreases but the hardness increases in the region of $x > 0.03$. The maximum strain and minimum hardness are 16.9% and 191 Hv, respectively. Note that the hardness of the FCC structured $FeMnNiCuCoSn_x$ alloys are softer than that of HEAs with BCC or FCC + BCC structures [12–16], because the closest packing planes in BCC structure possessing a smaller inter-planar spacing and a higher lattice friction to the dislocation motion are denser than that in FCC structure.

The fracture morphology of the tensile samples of the as-casted $FeMnNiCuCoSn_x$ alloys is demonstrated in Fig. 8. Fig. 8a shows dimpled rupture relying on the atomic fraction of Sn. The dimples are very deep and uniform and the edges have been pulled to threadiness after experiencing a strong plastic deformation. As x increases, the dimples become shallow, irregular and coarse, as shown in Fig. 8b and c corresponding to $FeMnNiCuCoSn_{0.08}$ and $FeMnNiCuCoSn_{0.1}$ alloys, respectively. The fracture surface of the $FeMnNiCuCoSn_{0.2}$ alloy displays a characteristic brittle, quazi-cleavage fracture morphology as shown in Fig. 8d.

Combining with the tensile experiment results, it is shown that $x = 0.03$ is the most appropriate Sn concentration for improving the plasticity of the as-cast $FeMnNiCuCoSn_x$ alloys, and the largest elongation strain can reach to 16.9%. Since the low ductility and brittleness limit the application of traditional HEAs, the as-cast $FeMnNiCuCoSn_x$ alloys are very useful for the future development and utilization of the HEAs.

4. Conclusions

The $FeMnNiCuCoSn_x$ HEAs show good plasticity and tensile strength at room temperature, depending on the effect of atomic fraction of Sn on microstructure. The $FeMnNiCuCoSn_x$ alloys are single FCC structures for x less than 0.05, whereas $Cu_{5.6}Sn$ intermetallic compound appears as x is higher than 0.05. Appropriate Sn is good at improving the plasticity of the as-casted $FeMnNiCuCoSn_x$ HEAs. When $0.03 < x < 0.05$, the alloys exhibit the high tensile strength and plasticity with the maxima of elongation strain and strength (16.9% and 476.9 MPa). Moreover, $x = 0.03$ is the most appropriate Sn concentration for improving the plasticity of the as-cast $FeMnNiCuCoSn_x$ alloys. Mn atoms dissolve into the FCC so-

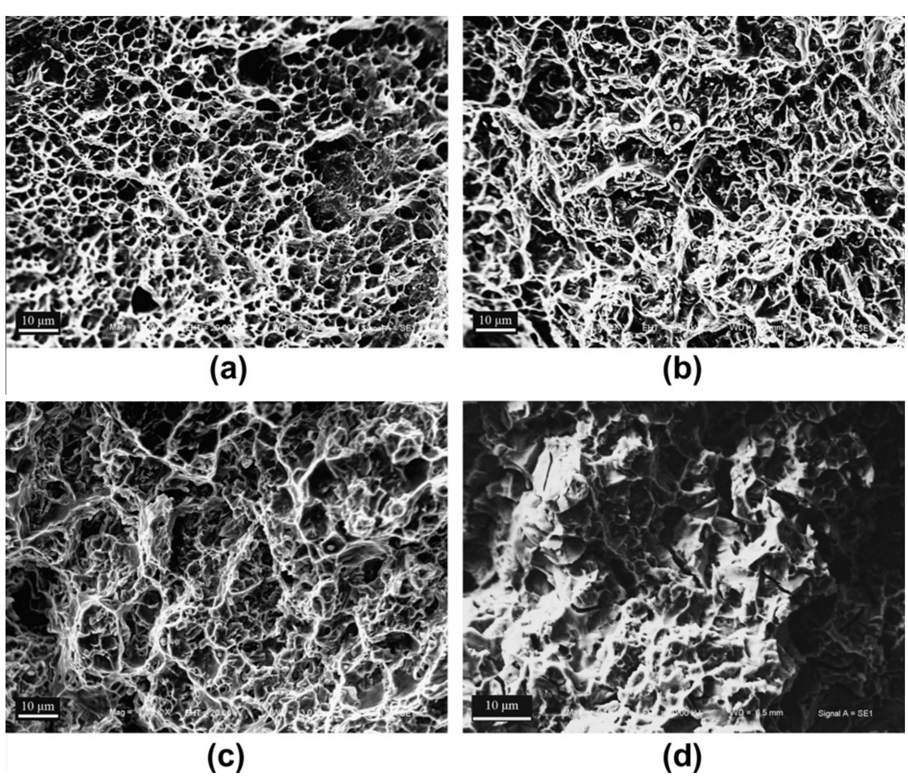

Fig. 8. SEM fracture morphology of the $FeMnNiCuCoSn_x$ alloys: (a) $x = 0.03$, (b) $x = 0.08$, (c) $x = 0.1$, and (d) $x = 0.2$.

lid solutions, and thus no Mn-included intermetallic compound appears.

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